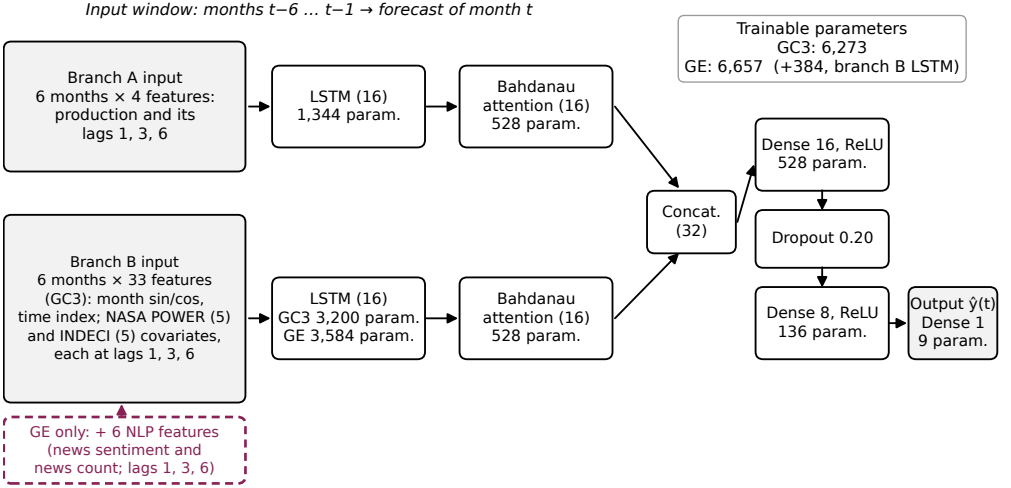
Input window: months t−6 … t−1 → forecast of month t
Branch A input
6 months × 4 features:
production and its
lags 1, 3, 6
LSTM (16)
1,344 param.
Bahdanau
attention (16)
528 param.
Trainable parameters
GC3: 6,273
GE: 6,657 (+384, branch B LSTM)
Branch B input
6 months × 33 features
(GC3): month sin/cos,
time index; NASA POWER (5)
and INDECI (5) covariates,
each at lags 1, 3, 6
LSTM (16)
GC3 3,200 param.
GE 3,584 param.
Bahdanau
attention (16)
528 param.
Concat.
(32)
Dense 16, ReLU
528 param.
Dropout 0.20
Dense 8, ReLU
136 param.
Output ŷ(t)
Dense 1
9 param.
GE only: + 6 NLP features
(news sentiment and
news count; lags 1, 3, 6)